TECHNICAL REPORT ON FRAUD DETECTION PREDICTION MODEL

Problem Definition and Business Significance

Bluewhale Bank is a mid-sized retail institution providing debit and credit card services across the UK and Europe, processing tens of thousands of transactions daily across mobile, online and point-of-sale channels. Despite existing rule-based controls, the bank continues to face rising fraud losses, high false-positive rates and increasing customer dissatisfaction due to incorrectly blocked legitimate transactions.

For this reason, the Bank aims to identify fraudulent transactions in near real time to prevent financial loss while preserving customer experience. To achieve this therefore, management has therefore approved the integration of Machine Learning (ML) solutions which is considered suitable as research has shown that Supervised ML models are widely applied in financial fraud detection to identify complex and evolving transactional patterns across multiple fraud domains (Compagnino et al., 2025; Hernandez Aros et al., 2024).

Prediction Target: The prediction target is **Fraud_Label**, a binary classification variable where 1 denotes a fraudulent transaction and 0 denotes a legitimate transaction. The objective of the model is not merely to produce a class label, but to estimate the probability that a given transaction is fraudulent.

Problem Significance to the Business: Fraud detection has significant implications for Bluewhale Bank. First, fraudulent transactions generate direct monetary losses through customer reimbursements, chargeback fees and investigative costs. Therefore, improvements in detection performance can yield substantial financial savings.

Second, false positives negatively affect customer experience by declining legitimate payments and increasing call-centre interactions, which may contribute to customer dissatisfaction and attrition. Effective fraud detection must minimise friction for genuine customers while maintaining robust security controls.

Third, regulatory frameworks require banks to demonstrate effective fraud monitoring and risk management. Weak detection systems expose the organisation to supervisory scrutiny, financial penalties and reputational damage.

Relevant Stakeholders: The implementation of a machine learning based fraud detection system affects multiple stakeholders within Bluewhale Bank, based on their interaction with the model outputs. The Fraud Operations Team are primary users of the model, using fraud probability scores to prioritise high-risk transactions, manage daily workloads and decide on blocks or refunds. The Risk & Compliance Department uses aggregated outputs to monitor fraud trends, calibrate decision thresholds and demonstrate regulatory oversight. Effective fraud monitoring requires continuous performance tracking due to evolving fraud behaviour and concept drift (Dal Pozzolo et al., 2014). Senior Management relies on performance metrics to track fraud losses, assess operational efficiency and guide strategic investment in fraud prevention. Data Science and Engineering teams

ensure technical deployment, monitor performance drift and maintain the ML pipeline. Customers are indirect stakeholders who benefit greatly from improved security and reduced unnecessary transaction declines.

DATA PREPARATION

The dataset was gotten from Kaggle

While the terms data processing and data preparation are often used interchangeably, data processing encompasses both data preparation and feature engineering (Njeri, 2022). Prior to model development, a structured data pre-processing was done to ensure that the dataset was complete, consistent and suitable for machine learning.

Data Understanding and Initial Checks: The dataset was inspected to understand its structure, variable types and target distribution. This involved reviewing feature types (continuous, binary, categorical, identifiers) and basic statistics to detect outliers and potential anomalies. The respective features were accessed for data type consistency, missing values and null values, however, there were no missing or duplicate values and no inconsistent daily count, which suggested a clean and coherent dataset. Timestamp conversion was done to access temporal fraud distribution on a monthly basis. This showed that fraud rates varied between approximately 31.2% and 33.1%, indicating a stable fraud prevalence over time.

Exploratory Data Analysis (EDA): This enables systematic examination of datasets to identify outliers and underlying relationships which provides a reliable foundation for subsequent feature transformation and modelling decisions (Dhummad, 2025). Univariate analysis was done using histograms and box plots to visualize skewness and outliers in the feature variables. Bivariate analysis measured continuous features against the target variable. However, categorical features were measured against target variable probability to remove size bias and directly measure risk as number of fraud cases per category can be misleading. For multivariate analysis, heatmap was used to explore correlation and detect redundancy.

Following the EDA, Feature relevance was assessed using filter-based methods (correlation analysis and Variance Inflation Factor) were used to identify multicollinearity and potential redundancy among variables.

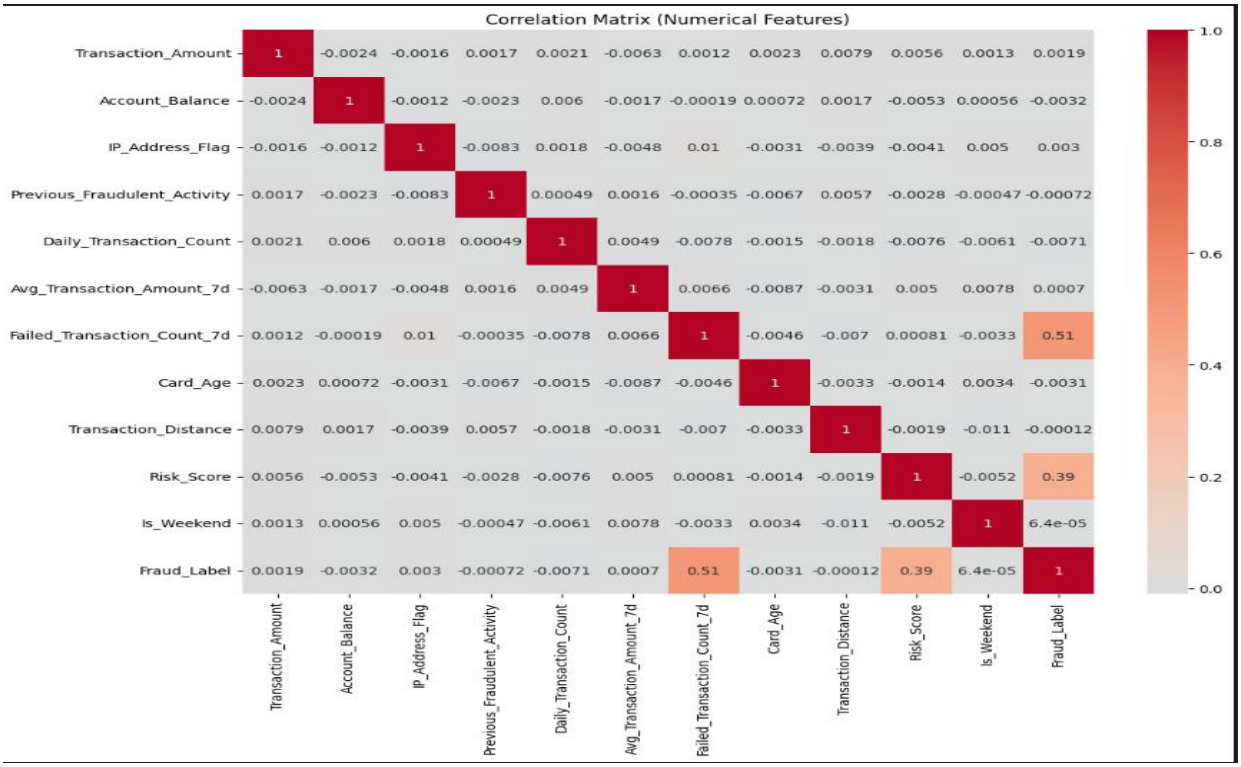

Fig 1. **Correlation Matrix**: Correlations with the target variable are generally low to moderate. The highest observed relationship is with Failed_Transaction_Count_7d, reinforcing the importance of recent transactional behaviour in fraud detection.

Feature Engineering: was done to transform the dataset from a collection of static transactional attributes to reflect awareness of customer behaviour and activity. In fraud detection, engineered behavioural and temporal features frequently outperform static magnitude-based variables because fraudulent activity is often pattern-driven rather than purely value-driven (Correa Bahnsen et al., 2016). This can be seen in the SHAP explainability below which shows that the model is mainly driven by recent behavioural patterns.

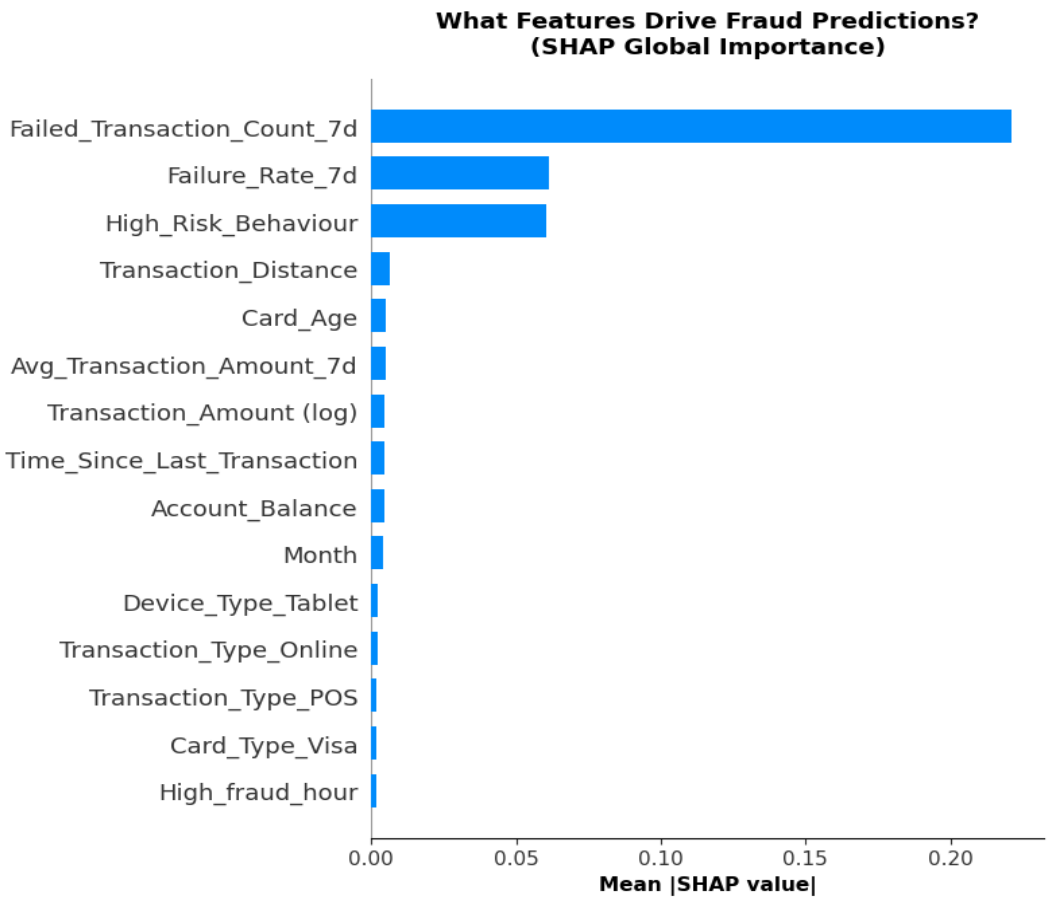

Fig 2. **SHAP Feature Importance:** confirms that recent behavioural features, particularly Failed_Transaction_Count_7d and Failure_Rate_7d, are the primary drivers of model predictions.

Furthermore, the timestamp variable was split into structured temporal components including hour, day, month and day of week to detect temporal fraud spikes. High Risk Behaviour was also engineered to capture suspicious behaviours and ratio features were also engineered to normalise transaction magnitude relative to account balance and historical spending behaviour.

Feature Scaling and Transformation: Standard scaling was performed on all continuous numeric features to ensure comparable influence across features, especially for Logistic Regression which is sensitive to scale. The EDA revealed significant skewness in the monetary variable and as such, log transformation was applied to reduce skewness and stabilize variance.

Encoding Categorical Variables: All categorical features were transformed using one-hot encoding to convert non-numeric classes into machine-readable binary vectors.

Handling Class Imbalance: The target variable exhibited a fraud prevalence of 32%, indicating an imbalanced classification problem which was addressed using class weights. This modifies the model's loss function by assigning greater weight to the minority class so that misclassification of those instances carries a larger penalty during training (Bakirarar & Elhan, 2023). This aligns better with cost-sensitive learning, since fraud detection penalizes costly errors.

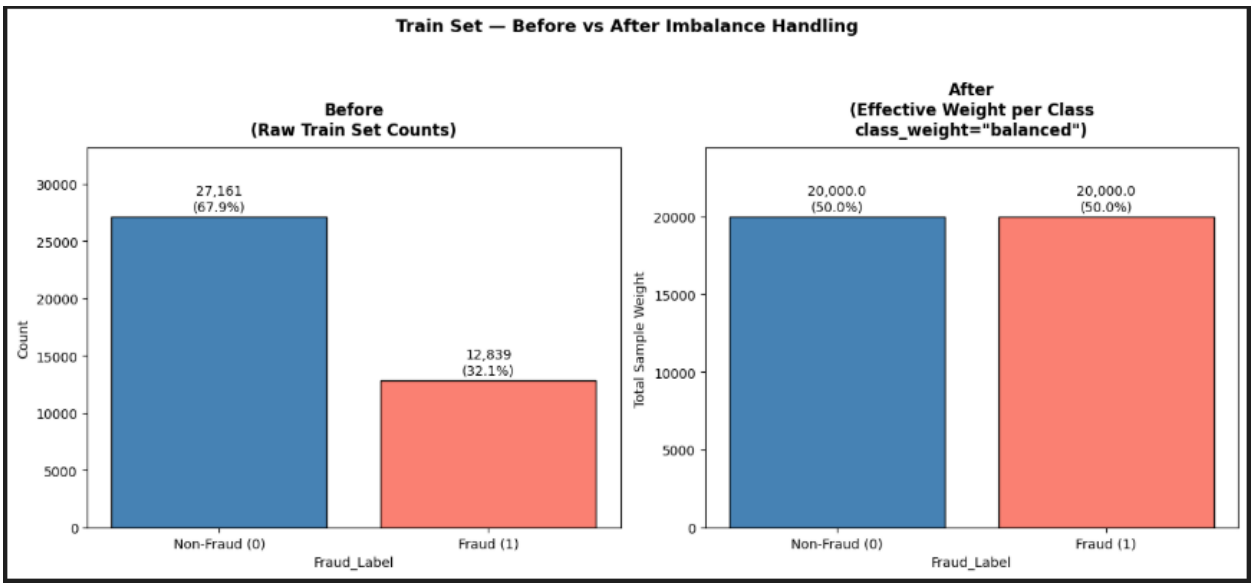

Fig 3. **Class Imbalance Handling:** From imbalanced to Balanced class following the application of class weights.

In addition, model pipeline was used to ensure consistent preprocessing.

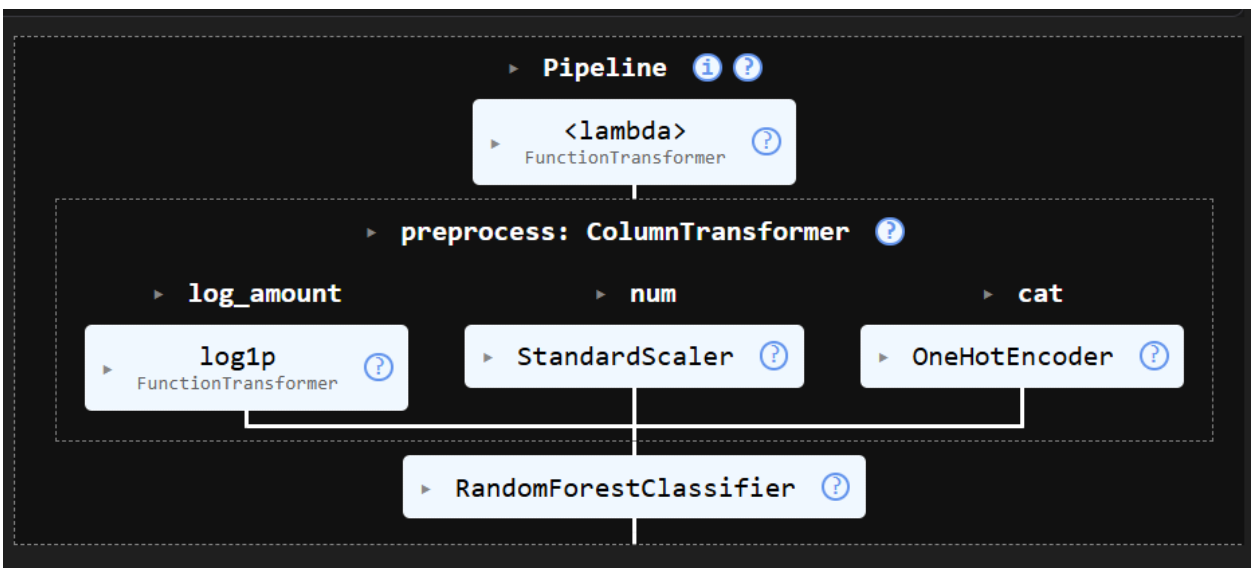

Fig 4. **End-to-End ML Pipeline**: this supports reproducibility, maintains data integrity during cross-validation, and enables seamless deployment of the trained model.

MODEL CHOICE AND JUSTIFICATION

Logistic Regression was used as a statistical baseline due to its transparency and ease of interpretation, with coefficients offering direct insight into feature influence. The model can be extended to handle multi-class problems, and once trained, it performs classification on new observations quickly and efficiently (Alenzi & O, 2020) However, its linear assumptions limit its ability to capture complex relationships.

Single decision trees are prone to overfitting and can be sensitive to variations in the training data. (Xuan et al., 2018). To address this Random Forest and XGBoost were selected. While Random Forest improves generalisation by aggregating multiple decision trees, reducing variance and overfitting (Breiman, 2001), XGBoost, enhances predictive accuracy by correcting errors which explains its strong performance especially on tabular datasets (Chen & Guestrin, 2016). Both models effectively captured non-linear interactions, making them the two best performing models.

Support Vector Machine (SVM) was included for its ability to construct optimal decision boundaries in high-dimensional spaces (Cortes et al., 1995), though it is less interpretable and computationally intensive.

Finally, a Dummy Classifier which served as the naïve baseline was used to establish the absolute minimum benchmark.

MODEL OPTIMISATION

Model performance was improved through targeted hyperparameter tuning of Random Forest, being the best performing model. Randomised Search method was implemented to identify best model configurations such as the number of trees, maximum depth, and minimum samples per split. This helped to reduce overfitting and control model complexity.

A stratified k-fold cross-validation strategy was implemented to ensure reliable model performance and reduce overfitting. This was chosen over standard cross-validation because it preserves class proportions and yields more reliable evaluation results (Widodo et al., 2022)

MODEL EVALUATION AND RESULTS

Accuracy alone is insufficient in evaluating model performance as it can be misleading. Therefore, precision, recall, F1-score, ROC-AUC and confusion metrics were used to show the proportion of correctly identified fraud cases, measure the ability to detect actual fraud and evaluate overall discriminatory power across thresholds (Saito & Rehmsmeier, 2015). Results indicated that ensemble models, particularly XGBoost and Random Forest, achieved higher ROC-AUC, demonstrating stronger capability in detecting fraudulent transactions.

```
Dummy
Mean CV ROC-AUC: 0.5

Logistic
Mean CV ROC-AUC: 0.8019548010523199

Random Forest
Mean CV ROC-AUC: 0.8774735843980949

XGBoost
Mean CV ROC-AUC: 0.874952253338608

SVM
Mean CV ROC-AUC: 0.8145005935902265
```

Fig 5. Cross-Validation Performance Comparison Across Models

Following optimization of the best performing model, the classification report indicates overall strong model performance, with an accuracy of 0.88 and an ROC-AUC score of 0.884, suggesting a good ability to discriminate between fraudulent and non-fraudulent transactions. For non-fraud, the model achieves high recall (0.95) and F1-score (0.91), indicating that most legitimate transactions are correctly identified, while for fraud class, precision remains relatively high (0.88), meaning flagged fraud cases are often correct. However, recall is lower (0.72), indicating that approximately 28% of fraudulent transactions are not detected.

```python
from sklearn.metrics import classification_report, roc_auc_score
print("Classification Report:")
print(classification_report(y_test, y_pred))
print("ROC-AUC Score:", roc_auc_score(y_test, y_proba))
```

Python

```
Classification Report:
              precision    recall  f1-score   support

           0       0.88      0.95      0.91      6772
           1       0.88      0.72      0.79      3228

    accuracy                           0.88     10000
   macro avg       0.88      0.84      0.85     10000
weighted avg       0.88      0.88      0.88     10000

ROC-AUC Score: 0.8840835706616134
```

Fig 6. Balanced Precision with Improved Fraud Recall

The Confusion metrics aligns with the classification report such that the high number of true negatives and relatively low false positives support the strong recall for non-fraud cases (0.95). For the fraud class, the presence of 904 false negatives explains the lower recall of 0.72, indicating

that some fraudulent transactions are still missed. The precision of 0.88 for fraud is consistent with the relatively low number of false positives (307), meaning most flagged transactions are indeed fraudulent.

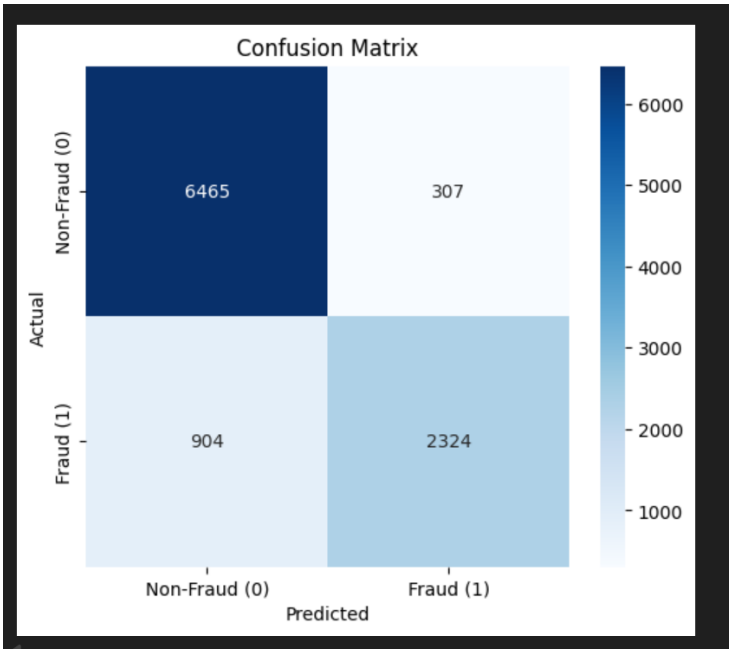

Fig 7. Confusion Matrix: Strong Non-Fraud Detection with Moderate Fraud Miss Rate.

While the model is effective at minimising false positives, it misses some fraud cases. In a business context, these false negatives represent direct financial risk, whereas false positives may lead to operational costs and customer friction.

	Model Version	CV ROC-AUC (Mean)	CV ROC-AUC (Std)	Test ROC-AUC	Test Accuracy	Test Precision	Test Recall	Test F1
0	Before Tuning	0.8775	0.0055	0.8850	0.8776	0.9434	0.6605	0.7770
1	After Tuning	0.8796	0.0040	0.8841	0.8789	0.8833	0.7200	0.7933

Fig 8. **Impact of Hyperparameter Tuning:** Stable ROC with Improved Recall.

ROC Curve Comparison: Random Forest and XGBoost outperform other models, indicating superior ability to capture non-linear fraud patterns.

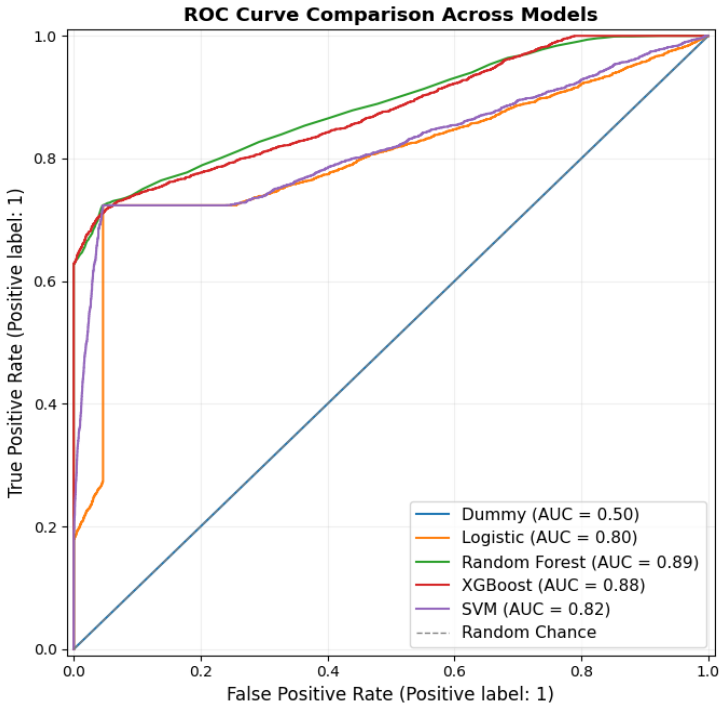

Fig 9. **ROC Curve Comparison**

Limitations and Future Improvements: Limitations of this model include potential concept drift over time due to evolving fraud patterns. Future improvements include threshold tuning, cost-sensitive learning, and ongoing model monitoring.
This study is based on a hypothetical financial institution (Bluewhale Bank) using a synthetic dataset. Consequently, reported model performance may overestimate real-world effectiveness, and findings should be interpreted within this limitation.

DEPLOYMENT CONSIDERATIONS

In practice, the trained model would be deployed as part of Bluewhale Bank's real-time transaction processing infrastructure. The modelling pipeline would be encapsulated within a secure RESTful API to be integrated directly with the bank's payment gateway, ensuring that each incoming transaction is scored before authorisation. End-to-end latency must remain within strict operational thresholds (typically milliseconds) to prevent disruption to customer experience. Deployment must also incorporate robust MLOps practices. This includes model versioning, reproducible artefact storage, environment management and audit logging to ensure traceability of predictions and model updates.

Random Forest is operationally suitable due to its strong predictive performance on structured tabular data and relatively efficient inference time.

ETHICAL LEGAL AND BIAS CONSIDERATIONS

In fraud detection, bias may arise indirectly through proxy variables such as location, device type or spending behaviour. This could lead to systematic over-flagging of certain customer segments, creating unfair treatment and reputational risk (Mehrabi et al., 2022). Accordingly, fairness evaluation metrics should be incorporated into model governance.

From a legal perspective, deployment must comply with data protection regulations such as the UK GDPR. Automated decision-making that significantly affects individuals requires transparency and the ability to provide meaningful explanations (Goodman & Flaxman, 2017). A robust deployment strategy should include:

- Model documentation
- Bias and fairness audits
- Transparent threshold-setting aligned with business risk tolerance
- Human-in-the-loop review for high-risk or borderline cases
- Ongoing monitoring for drift and unintended discrimination

Responsible Artificial Intelligence deployment is not a one-off activity but an ongoing governance process integrating technical, legal and ethical oversight.

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